

Gravitational waves from first-order phase transitions[⌚]

Lecture 3: wall velocity and spectra

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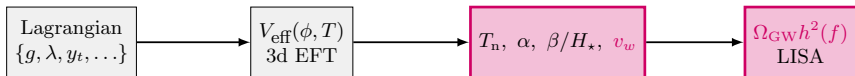
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[⌚] Ekstedt, Gould, Hirvonen *et al.*, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04** (2025) 101 [2411.04970]

Supported by the SNSF under grant PZ00P2-215997.

Where we are



Lecture 1: the effective potential and the infrared problem.

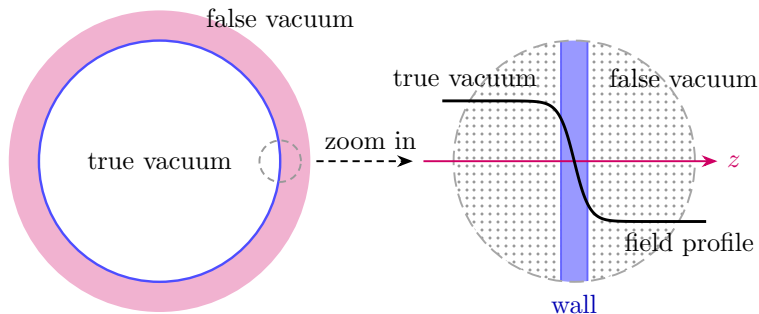
Lecture 2: the 3d EFT, the Standard Model and beyond, and the nucleation temperature T_n .

Today the bubble *grows*, *collides* and *radiates*:

- (a) the wall velocity v_w , at the interface between particle physics and hydrodynamics;
- (b) the gravitational-wave spectrum it produces.

What is the wall velocity?

The bubble wall



The *bubble wall* is the region where the field(s) interpolate between the high- and the low-temperature vacuum.

In the wall frame, the plasma streams through it with velocity v_w .¹

¹Figure after Ai, Laurent, van de Vis, *Bounds on the bubble wall velocity*, JHEP **02** (2025) 119 [2411.13641]

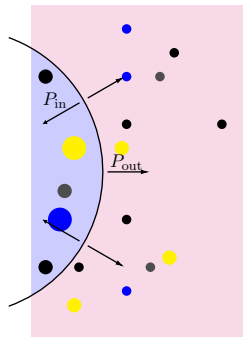
Forces on the wall

Outward, P_{out} :

the released vacuum energy accelerates the wall.

Inward, P_{in} :

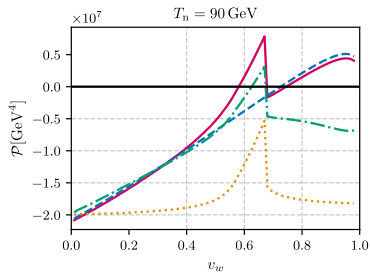
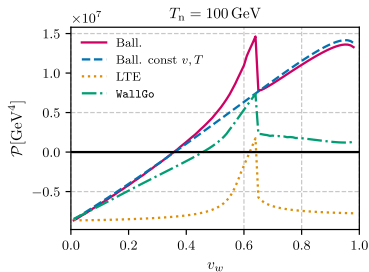
- ▷ *friction*: particles reflect off the wall, or gain mass on entering it;
- ▷ *hydrodynamic backreaction*: the plasma in front is heated and compressed.



Terminal velocity: $|P_{\text{out}}| = |P_{\text{in}}|$. If it is reached early, $t_{\text{form}} \ll t_{\text{const } v_w} \ll t_{\text{coll}}$, a single number v_w describes the whole expansion.

Runaway: if P_{in} never catches up, the wall accelerates until the bubbles collide.

The terminal velocity is a zero crossing



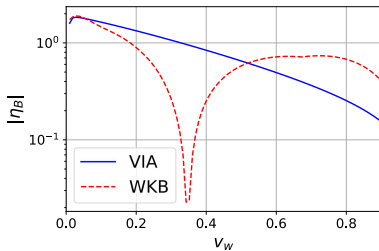
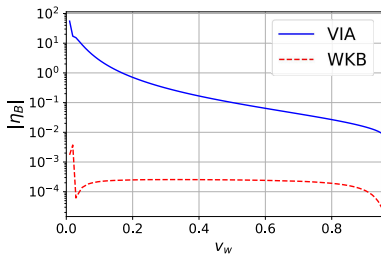
Net pressure $\mathcal{P}(v_w)$ on the wall in the xSM, at $T_n = 100$ GeV (left) and 90 GeV (right).²

- ▷ $\mathcal{P}(v_w) = 0$ defines v_w .
- ▷ The jump is at the Jouguet velocity v_j (hybrid $\rightarrow$ detonation).

² Ekstedt, Gould, Hirvonen *et al.*, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP 04 (2025) 101 [2411.04970]

Why do we want to know v_w ?

Electroweak baryogenesis depends on the wall velocity



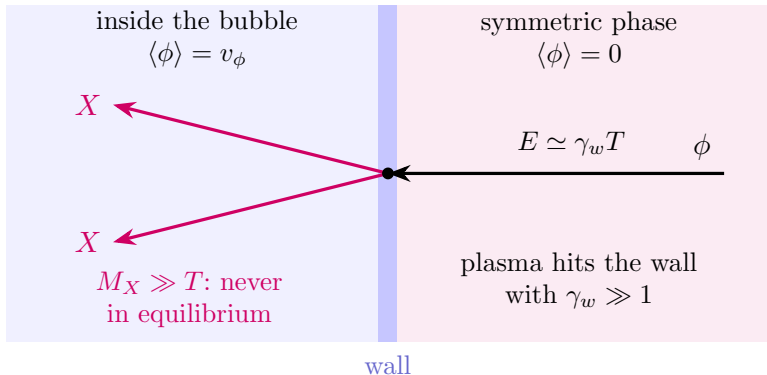
Baryon asymmetry from CP violation in a lepton Yukawa (left) and from c - t flavor mixing (right).³

- ▷ It changes by orders of magnitude with v_w , not monotonically.
- ▷ The transport formalisms VIA and WKB disagree even on the shape.

³ Cline, Laurent, *Electroweak baryogenesis from light fermion sources: A critical study*, Phys. Rev. D **104** (2021) 083507 [2108.04249]

Dark matter production depends on the wall velocity

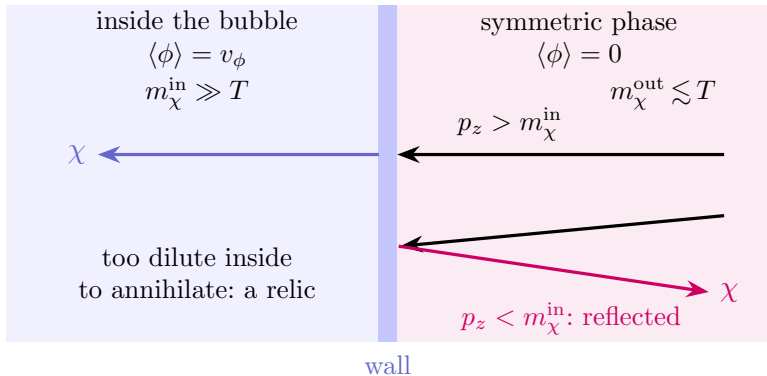
Fast walls, $\gamma_w \gg 1$: in the wall frame the plasma arrives with $E \simeq \gamma_w T$, and $\phi \rightarrow XX$ across the wall produces dark matter much heavier than T .⁴



⁴ Azatov, Vanvlasselaer, Yin, *Dark Matter production from relativistic bubble walls*, JHEP **03** (2021) 288 [2101.05721]

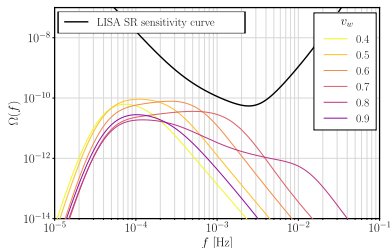
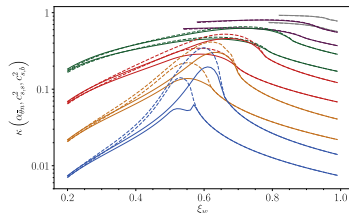
Dark matter production depends on the wall velocity

Slow walls: χ is light outside and heavy inside. Only χ with $p_z > m_\chi^{\text{in}}$ get in, the rest is reflected: “filtered” dark matter.⁵



⁵ Baker, Kopp, Long, *Filtered Dark Matter at a First Order Phase Transition*, Phys. Rev. Lett. **125** (2020) 151102 [1912.02830]

And the gravitational-wave spectrum depends on it too



Left: efficiency κ of converting vacuum energy into fluid motion.

Right: spectra for $0.4 \leq v_w \leq 0.9$ at fixed $\alpha = 0.2$, $r_\star = 0.1$, $T_n = 100$ GeV.⁶

Changing only v_w shifts the peak, and the tail at 10^{-2} Hz by orders of magnitude.

⁶ Gowling, Hindmarsh, *Observational prospects for phase transitions at LISA: Fisher matrix analysis*, JCAP **10** (2021) 039 [2106.05984]

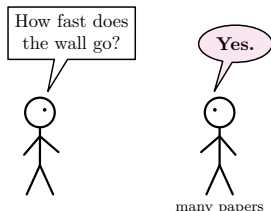
What makes v_w hard

- ▷ **Coupled:** scalar equation of motion, hydrodynamics and a Boltzmann equation with nine-dimensional collision integrals, solved together.
- ▷ **Model-dependent:** particle content, masses *and* matrix elements.

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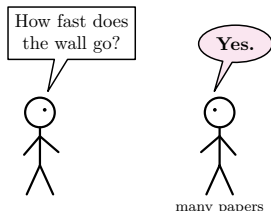
⇒ Many papers in the literature simply set $v_w = 1$.



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Rest of this lecture: what is involved, and what ~~Wall~~Go⁷ does.

⁷ Ekstedt, Gould, Hirvonen *et al.*, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04** (2025) 101 [2411.04970]

Wall velocity: theory

Assumptions

- ▷ We compute v_w while the bubble still expands *in isolation*, $t_{\text{formation}} \ll t_{\text{const } v_w} \ll t_{\text{collision}}$.
- ▷ The bubbles are much smaller than the horizon, $R_{\text{bubble}}H \ll 1$, so we work in Minkowski spacetime and treat the wall as planar.
- ▷ The plasma far in front of the bubble is at the nucleation temperature T_n , *the output of lecture 2*.
- ▷ The theory is weakly coupled, so a quasiparticle description and a Boltzmann equation make sense.

The scalar field equation of motion

$$\partial^2 \phi_i + \underbrace{\frac{\partial V_{\text{eff}}(\phi, T)}{\partial \phi_i}}_{\text{driving force + backreaction}} + \underbrace{\sum_a \frac{\partial m_a^2}{\partial \phi_i} \int_{\mathbf{p}} \frac{\delta f^a(p^\mu, \xi)}{2E}}_{\text{friction}} = 0$$

- ▷ V_{eff} : the effective potential including thermal corrections (lecture 1).
- ▷ δf^a : deviation of the distribution of species a from equilibrium.
- ▷ The friction is dominated by *heavy* species, with large $\partial m_a^2 / \partial \phi$: the top quark, the W and Z .
- ▷ $\xi = -\bar{u}_w^\mu x_\mu$ is the distance from the wall, with $\bar{u}_w^\mu = \gamma_w(v_w, 0, 0, 1)$.

Balance of forces

Multiply the equation of motion by $\phi' = d\phi/dz$ and integrate across the wall: each force density becomes a force per area, the *pressure*. At constant speed it vanishes:⁸

$$\begin{aligned}\mathcal{P} &= \int dz \phi' [\text{EOM}] \\ &= \underbrace{-\int dz \phi' \phi''}_{\text{total derivative} = 0} + \int d\phi \frac{\partial V_{\text{eff}}}{\partial \phi} + \sum_a \int d\phi \frac{\partial m_a^2}{\partial \phi} \int_{\mathbf{p}} \frac{\delta f^a}{2E} \\ &= \underbrace{\Delta V_{\text{eff}}}_{\text{driving}} - \underbrace{\int dz \frac{\partial V_{\text{eff}}}{\partial T} \frac{dT}{dz}}_{\text{backreaction}} + \underbrace{\sum_a \int dz \frac{dm_a^2}{dz} \int_{\mathbf{p}} \frac{\delta f^a}{2E}}_{\text{friction}} = 0.\end{aligned}$$

⁸ Ai, Garbrecht, Tamarit, *Bubble wall velocities in local equilibrium*, JCAP **03** (2022) 015 [2109.13710]

Temperature and fluid profile

Energy-momentum tensor of the (perfect) fluid plus scalar field:

$$T_{\mu\nu} = \underbrace{w}_{\text{enthalpy}} u_\mu u_\nu - \underbrace{p}_{=-V_{\text{eff}}(\phi, T)} g_{\mu\nu} + \partial_\mu \phi_i \partial_\nu \phi_i - g_{\mu\nu} \left(\frac{1}{2} \partial_\sigma \phi_i \partial^\sigma \phi_i \right),$$

with $w = T \, dp/dT$.

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with $w = T \, dp/dT$. For a planar wall moving in z , $\partial_\mu T^{\mu\nu} = 0$ gives two first integrals,

$$\begin{aligned} T^{30} &= w \gamma_{\text{pl}}^2 v_{\text{pl}} + T_{\text{out}}^{30} = c_1, \\ T^{33} &= \frac{1}{2} (\partial_z \phi_i)^2 - V_{\text{eff}}(\phi, T) + w \gamma_{\text{pl}}^2 v_{\text{pl}}^2 + T_{\text{out}}^{33} = c_2, \end{aligned}$$

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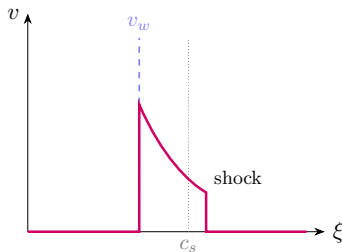
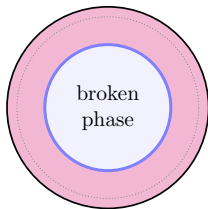
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with c_1, c_2 fixed by the hydrodynamic solution *far* from the wall.

Bubble growth as relativistic combustion

The wall is a *combustion front*: it turns the high- T phase into the low- T phase and releases latent heat. Away from it, $\partial_\mu T^{\mu\nu} = 0$ has self-similar solutions $v(\xi = r/t)$.⁹

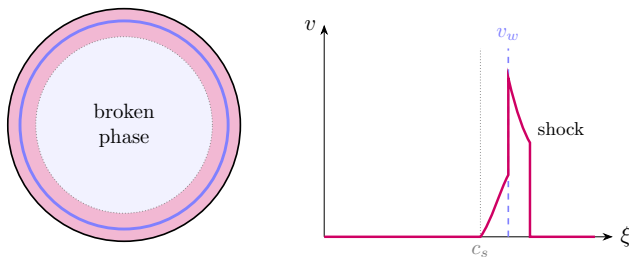


Deflagration, $v_w < c_s$: fluid pushed ahead, up to a shock.

⁹ Steinhardt, *Relativistic Detonation Waves and Bubble Growth in False Vacuum Decay*, Phys. Rev. D **25** (1982) 2074; Espinosa, Konstandin, No, Servant, *Energy Budget of Cosmological First-order Phase Transitions*, JCAP **06** (2010) 028 [1004.4187]

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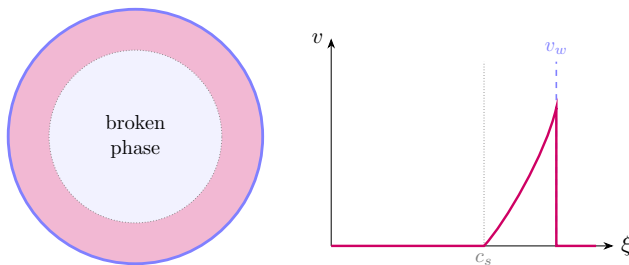


Hybrid, $c_s < v_w < v_J$: shock ahead, rarefaction behind.

⁹ Steinhardt, *Relativistic Detonation Waves and Bubble Growth in False Vacuum Decay*, Phys. Rev. D **25** (1982) 2074; Espinosa, Konstandin, No, Servant, *Energy Budget of Cosmological First-order Phase Transitions*, JCAP **06** (2010) 028 [1004.4187]

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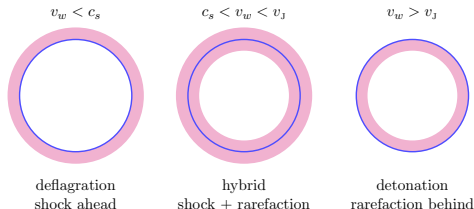


Detonation, $v_w > v_J$: fluid ahead at rest, rarefaction behind.

⁹ Steinhardt, *Relativistic Detonation Waves and Bubble Growth in False Vacuum Decay*, Phys. Rev. D **25** (1982) 2074; Espinosa, Konstandin, No, Servant, *Energy Budget of Cosmological First-order Phase Transitions*, JCAP **06** (2010) 028 [1004.4187]

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Backreaction: the heated fluid ahead pushes back. It grows with v_w for deflagrations and hybrids and falls for detonations, so $\mathcal{P}(v_w)$ jumps at v_J .

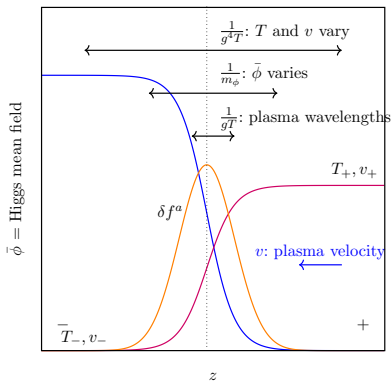
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Matching the scales

Three scales inside the wall:

- ▷ $1/m_\phi$: width of the field profile $\phi(z)$;
- ▷ $1/(gT)$: wavelength of the plasma particles that carry δf^a ;
- ▷ $1/(g^4 T)$: distance over which T and v change.

On hydrodynamic scales the wall is a *discontinuity* in T and v . Given T_n and v_w , the matching conditions fix $T_\pm$, $v_\pm$ and hence c_1, c_2 .



Obtaining ϕ_i , v , T and v_w

Assume for a moment that we know δf^a . Then:

- ▷ solve for ϕ_i, v, T at *fixed* v_w ;
- ▷ evaluate the pressure $\mathcal{P}$ on the wall;
- ▷ $\mathcal{P}(v_w) = 0$ gives the wall velocity.

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Often (also in WallGo) a tanh ansatz for the fields:

$$\phi_i(z) = v_{\text{HT},i} + \frac{v_{\text{LT},i} - v_{\text{HT},i}}{2} \left[1 - \tanh\left(\frac{z}{L_i} + \delta_i\right) \right]$$

$v_{\text{HT},i}, v_{\text{LT},i}$: field values outside and inside the bubble;
 L_i, δ_i : wall width and offset, fixed by moments of the EOM.

Problems

Problem 1: Derive the two first integrals $T^{30} = c_1$, $T^{33} = c_2$ from $\partial_\mu T^{\mu\nu} = 0$ for a stationary planar wall, and show that they reduce to the usual matching conditions $w_+ \gamma_+^2 v_+ = w_- \gamma_-^2 v_-$ when $\partial_z \phi = 0$.

Problem 2: Show that in local thermal equilibrium entropy conservation, $\partial_\mu (su^\mu) = 0$, provides a third matching condition, and that it determines v_w without any collision integral.

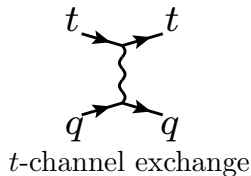
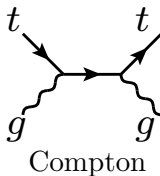
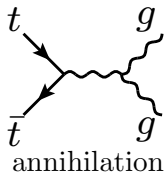
Problem 3: Using the tanh ansatz, compute $\int dz (d\phi/dz)^2$ and show that the surface tension scales as $\sigma \propto (v_{\text{LT}} - v_{\text{HT}})^2 / L_\phi$.

The out-of-equilibrium contribution

The Boltzmann equation

$$\underbrace{\left(p^\mu \partial_\mu + \frac{1}{2} \nabla m_a^2 \cdot \nabla_{\mathbf{p}}\right)}_{\text{force term}} f^a(\mathbf{p}, \xi) = - \underbrace{\mathcal{C}_a[\mathbf{f}]}_{\text{collisions}}$$

The collision term is a **nine-dimensional** integral over the distributions of the four species in each $2 \rightarrow 2$ process. For the top quark, for example:



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The collision term is a **nine-dimensional** integral over the distributions of the four species in each $2 \rightarrow 2$ process. Linearize around equilibrium, $f^a = f_{\text{eq}}^a + \delta f^a$ with $\mathcal{C}[f_{\text{eq}}] = 0$:

$$\left(-p_\mu \bar{u}_w^\mu \partial_\xi - \frac{1}{2} \partial_\xi (m_a^2) \bar{u}_w^\mu \partial_{p^\mu}\right) \delta f^a = -\mathcal{C}_{ab}^{\text{lin}}[\delta f^b] + \mathcal{S}_a.$$

Source $\mathcal{S}_a$: the same operator acting on f_{eq}^a .

Off-diagonal $\mathcal{C}_{ab}^{\text{lin}}$: mixing between species, e.g. top and W .

Strategy (a): taking moments

Truncate the perturbation as $f = f_{\text{eq}}(E + \delta)$ with $\delta = -[\mu + (E/T)\delta T + p_z v]$.¹⁰ Taking the moments

$$\int \frac{d^3p}{(2\pi)^3}, \quad \int \frac{E d^3p}{(2\pi)^3}, \quad \int \frac{p_z d^3p}{(2\pi)^3}$$

gives three coupled ODEs for $(\mu, \delta T, v)$: particle number, energy and momentum conservation.

Advantages.

- ▷ numerically cheap;
- ▷ collision terms reduce to a few numbers, e.g. for top quarks $\Gamma_{\mu 1f} = 0.00899 T$.

Disadvantages.

- ▷ spurious singularity at $v_w = c_s$;
- ▷ convergence in moments not controlled;
- ▷ no species mixing.

¹⁰ Moore, Prokopec, *How fast can the wall move? A Study of the electroweak phase transition dynamics*, Phys. Rev. D **52** (1995) 7182 [hep-ph/9506475]

Strategy (b): expanding in polynomials

Expand in Chebyshev polynomials in rescaled coordinates:¹¹

$$\delta f^a(\chi, \rho_z, \rho_{\parallel}) = \sum_{i,j,k} \delta f_{ijk}^a \bar{T}_i(\chi) \bar{T}_j(\rho_z) \tilde{T}_k(\rho_{\parallel}).$$

The Boltzmann equation becomes a linear *matrix* equation.

Advantages.

- ▷ exponential convergence;
- ▷ no spurious singularity;
- ▷ species mixing included.

Disadvantages.

- ▷ numerically more expensive;
- ▷ the coefficients have no individual meaning.

¹¹ Laurent, Cline, *First principles determination of bubble wall velocity*, Phys. Rev. D **106** (2022) 023501 [2204.13120]

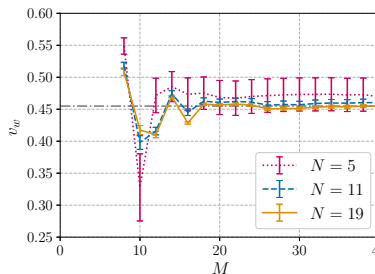
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Convergence of v_w
with the number M of
position points:
error $\propto e^{-0.15M}$.



¹¹ Laurent, Cline, *First principles determination of bubble wall velocity*, Phys. Rev. D **106** (2022) 023501 [2204.13120]

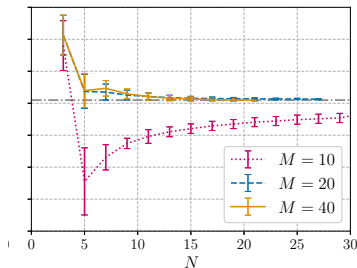
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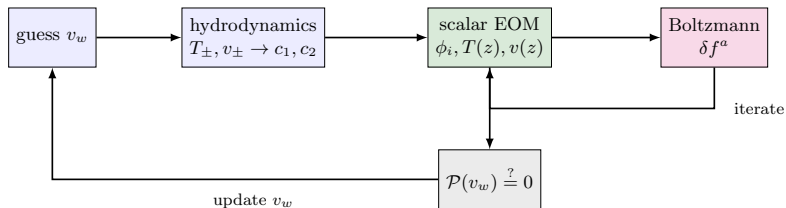
The Boltzmann equation becomes a linear *matrix* equation.

Convergence of v_w
with the number N of
momentum points:
error $\propto e^{-0.36N}$.



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The full iteration



In practice:

1. solve for ϕ_i, v, T at fixed v_w ;
2. compute δf^a on that background, and feed it back;
3. iterate 1–2 until convergence;
4. scan over v_w to find the zero of $\mathcal{P}$.

Results for v_w

Two limits of the collision term

Local thermal equilibrium (LTE), $C_a \rightarrow \infty$.¹² Then $\delta f^a \rightarrow 0$, and the friction drops out:

$$\partial^2 \phi_i + \frac{\partial V_{\text{eff}}(\phi, T)}{\partial \phi_i} + \underbrace{\sum_a \frac{\partial m_a^2}{\partial \phi_i} \int_{\mathbf{p}} \frac{\delta f^a}{2E}}_{\rightarrow 0} = 0.$$

Entropy is conserved across the wall, $\partial_\mu(su^\mu) = 0$: a *third* matching condition, which fixes v_w from hydrodynamics alone.

Friction can only slow the wall further, so LTE gives an **upper bound** on v_w .

¹² Ai, Laurent, van de Vis, *Model-independent bubble wall velocities in local thermal equilibrium*, JCAP **07** (2023) 002 [2303.10171]

Two limits of the collision term

Ballistic limit, $\mathcal{C}_a \rightarrow 0$.¹³ Particles cross the wall without scattering:

$$\left(p^\mu \partial_\mu + \frac{1}{2} \nabla m_a^2 \cdot \nabla_{\mathbf{p}} \right) f^a = \underbrace{-\mathcal{C}_a[\mathbf{f}]}_{\rightarrow 0} .$$

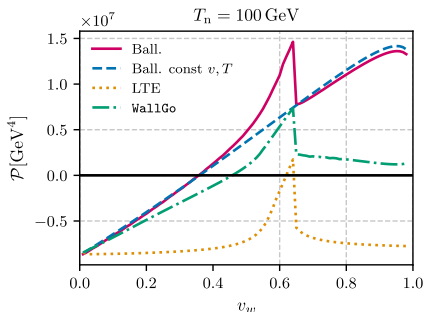
E and $\mathbf{p}_\perp$ are conserved, so f^a is constant along each trajectory; particles with $p_z^2 < \Delta m_a^2$ are reflected.

Solvable analytically, and a **lower bound** on v_w .

¹³ Ai, Laurent, van de Vis, *Bounds on the bubble wall velocity*, JHEP **02** (2025) 119 [2411.13641]

Pinching v_w between two limits

The two limits bracket the full result: **LTE** gives an **upper** bound on v_w , the **ballistic** limit a **lower** bound.¹⁴



xSM, $T_n = 100 \text{ GeV}$: $v_w^b < v_w^{\text{full}} < v_w^{\text{LTE}}$. Where the limits are far apart, the Boltzmann equation must be solved.

¹⁴ Ekstedt, Gould, Hirvonen *et al.*, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04** (2025) 101 [2411.04970]

Gravitational-wave spectra

Why gravitational waves survive

Question. Photons were rescattered until the CMB formed. Why do gravitational waves still remember the transition?

Rate. A plasma emits gravitons only at order $1/M_{\text{Pl}}^2$ ($\alpha_s = g^2/4\pi$):

$$\frac{de_{\text{GW}}}{dt d \ln k} \sim \frac{\alpha_s}{M_{\text{Pl}}^2} k^4 n_{\text{B}}(k) \hat{\Phi}(k, T) \stackrel{k \sim T}{\sim} \frac{\alpha_s T^7}{M_{\text{Pl}}^2}.$$

Divided by the energy density $\sim T^4$, this is the rate $\Gamma_{\text{GW}} \sim \alpha_s T^3/M_{\text{Pl}}^2$ at which gravitons would approach equilibrium. Compared with the expansion, $H \sim T^2/M_{\text{Pl}}$:

$$\frac{\Gamma_{\text{GW}}}{H} \sim \frac{\alpha_s T^3/M_{\text{Pl}}^2}{T^2/M_{\text{Pl}}} = \frac{\alpha_s T}{M_{\text{Pl}}}.$$

Why gravitational waves survive

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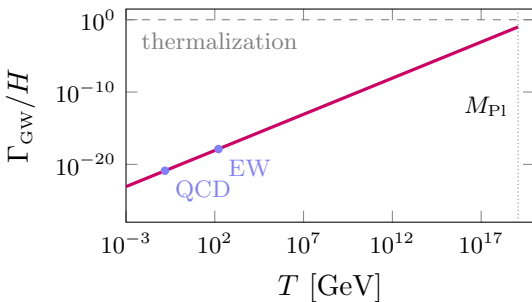
Rate. A plasma emits gravitons only at order $1/M_{\text{Pl}}^2$ ($\alpha_s = g^2/4\pi$):

$$\frac{d\epsilon_{\text{GW}}}{dt d\ln k} \sim \frac{\alpha_s}{M_{\text{Pl}}^2} k^4 n_{\text{B}}(k) \hat{\Phi}(k, T) \stackrel{k \approx T}{\sim} \frac{\alpha_s T^7}{M_{\text{Pl}}^2}.$$

$\alpha_s \approx 0.1$:

$\Gamma_{\text{GW}}/H \sim 10^{-18}$ at the electroweak scale, still < 1 at M_{Pl} .

So gravitational waves *free-stream* from the moment they are made.



Why gravitational waves survive

Question. Photons were rescattered until the CMB formed. Why do gravitational waves still remember the transition?

Source. The rate is a two-point function of $T^{\mu\nu}$:

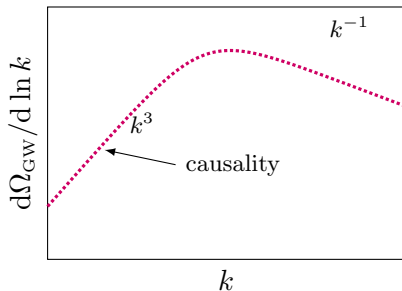
$$\frac{de_{\text{GW}}}{dt d^3\mathbf{k}} = \underbrace{\frac{2}{\pi^2 M_{\text{Pl}}^2}}_{\text{coupling } \propto G} \underbrace{\frac{1}{(2\pi)^3} \sum_{\lambda} \varepsilon_{\mu\nu}^{\text{TT}*} \varepsilon_{\rho\sigma}^{\text{TT}}}_{\text{2 polarizations}} \underbrace{\int_X e^{iK \cdot X} \langle \hat{T}^{\mu\nu}(0) \hat{T}^{\rho\sigma}(X) \rangle}_{\text{fluctuations of } T^{\mu\nu}} .$$

In equilibrium: thermal fluctuations of $T^{\mu\nu}$ at scale $1/T$. Tiny rate.

During the transition: the bubbles set the fluid in bulk motion, $T^{ij} \supset w\gamma^2 v^i v^j$, a far larger stress than the thermal fluctuations. Once bubbles collide, its transverse-traceless part is nonzero and radiates at $k \sim 1/\ell_{\text{B}}$.

Walls, sound waves, turbulence: the spectra

Walls, envelope approximation.¹⁵ Only the thin scalar-field shells radiate, and they vanish where bubbles overlap.
Spectrum: k^3 at small k (causality), k^{-1} at large k .

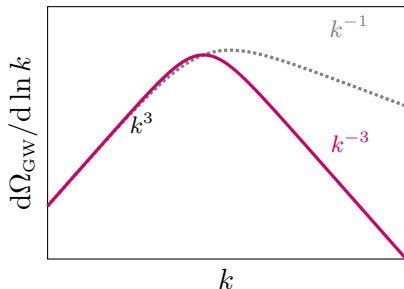
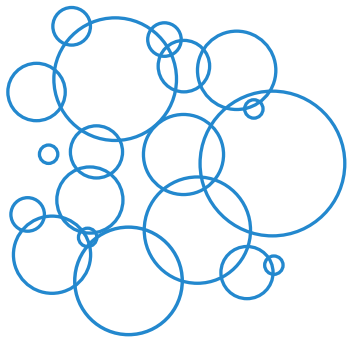


¹⁵ Kosowsky, Turner, *Gravitational radiation from colliding vacuum bubbles: envelope approximation to many bubble collisions*, Phys. Rev. D **47** (1993) 4372 [astro-ph/9211004]; Weir, *Revisiting the envelope approximation: gravitational waves from bubble collisions*, Phys. Rev. D **93** (2016) 124037 [1604.08429]

Walls, sound waves, turbulence: the spectra

Walls, full collisions.¹⁶ Keeping the collided walls steepens the tail from k^{-1} to k^{-3} .

The envelope result (gray) overestimates the high- k power.

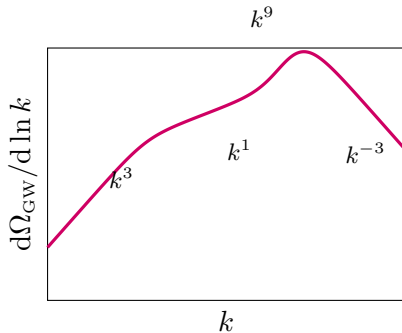
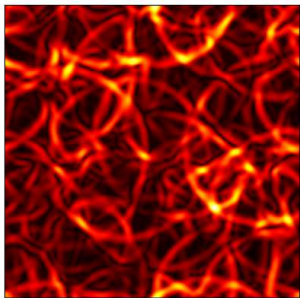


¹⁶ Cutting, Hindmarsh, Weir, *Gravitational waves from vacuum first-order phase transitions: from the envelope to the lattice*, Phys. Rev. D **97** (2018) 123513 [1802.05712]; Lewicki, Vaskonen, *Gravitational waves from colliding vacuum bubbles in gauge theories*, Eur. Phys. J. C **81** (2021) 437 [2012.07826]

Walls, sound waves, turbulence: the spectra

Sound waves.¹⁷ The fluid shells keep ringing long after the walls have gone: the dominant source for weak and intermediate transitions.

Four regimes: $k^3 \rightarrow k^1 \rightarrow k^9 \rightarrow k^{-3}$.

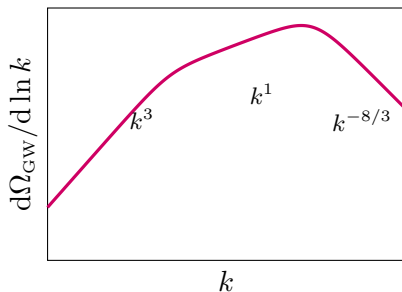
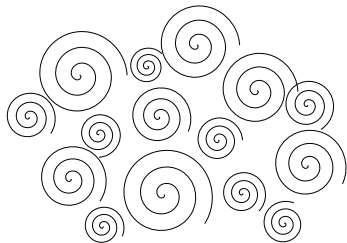


¹⁷ Hindmarsh, Huber, Rummukainen, Weir, *Gravitational waves from the sound of a first order phase transition*, Phys. Rev. Lett. **112** (2014) 041301 [1304.2433]; Hindmarsh, *Sound shell model for acoustic gravitational wave production at a first-order phase transition in the early Universe*, Phys. Rev. Lett. **120** (2018) 071301 [1608.04735]

Walls, sound waves, turbulence: the spectra

Vortical turbulence.¹⁸ In strong transitions the fluid motion turns non-linear and the kinetic energy cascades to small scales (Kolmogorov).

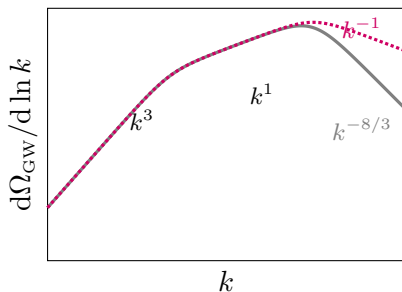
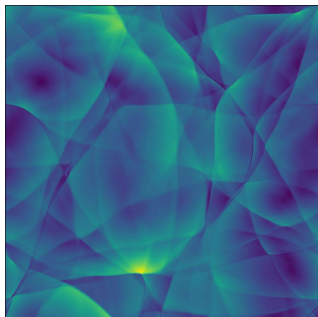
The tail becomes $k^{-8/3}$.



¹⁸ Caprini, Durrer, Servant, *The stochastic gravitational wave background from turbulence and magnetic fields generated by a first-order phase transition*, JCAP **12** (2009) 024 [0909.0622]; Roper Pol, Mandal, Brandenburg, Kahnishvili, Kosowsky, *Numerical simulations of gravitational waves from early-universe turbulence*, Phys. Rev. D **102** (2020) 083512 [1903.08585]

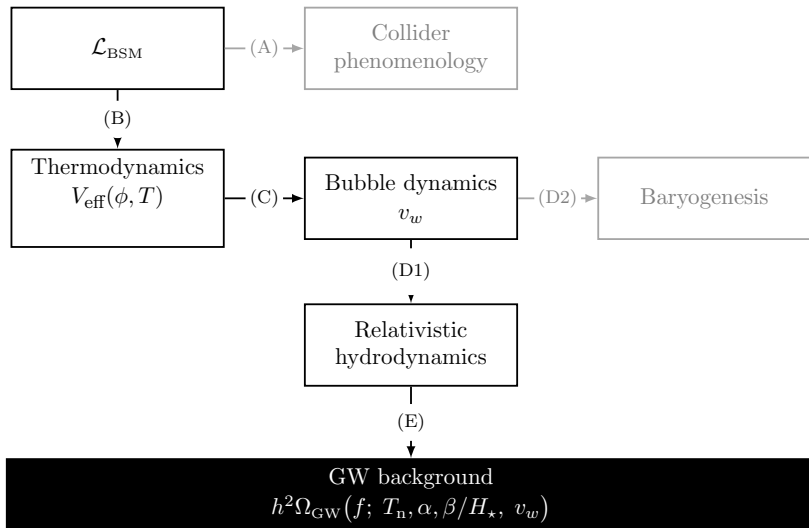
Walls, sound waves, turbulence: the spectra

Acoustic turbulence.¹⁹ The sound waves steepen into *shocks*, which decay differently from vortical turbulence. The tail flattens to k^{-1} (dotted), against $k^{-8/3}$ (gray).

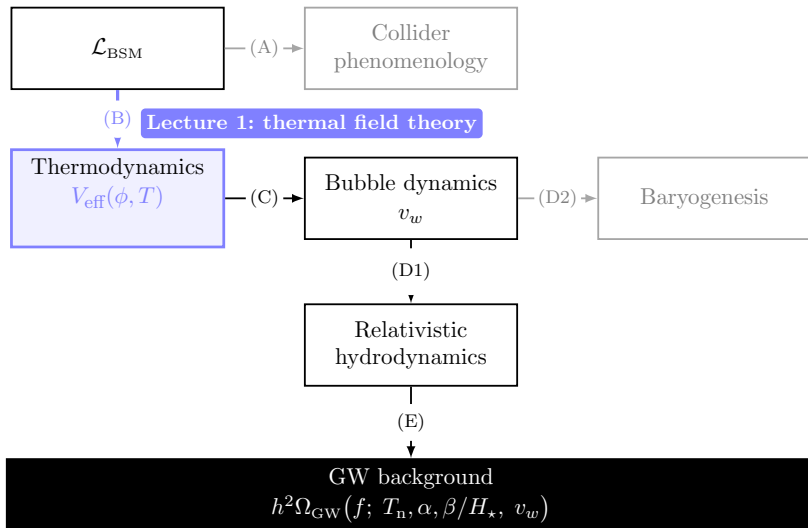


¹⁹ Dahl, Hindmarsh, Rummukainen, Weir, *Decay of acoustic turbulence in two dimensions and implications for cosmological gravitational waves*, Phys. Rev. D **106** (2022) 063511 [2112.12013]

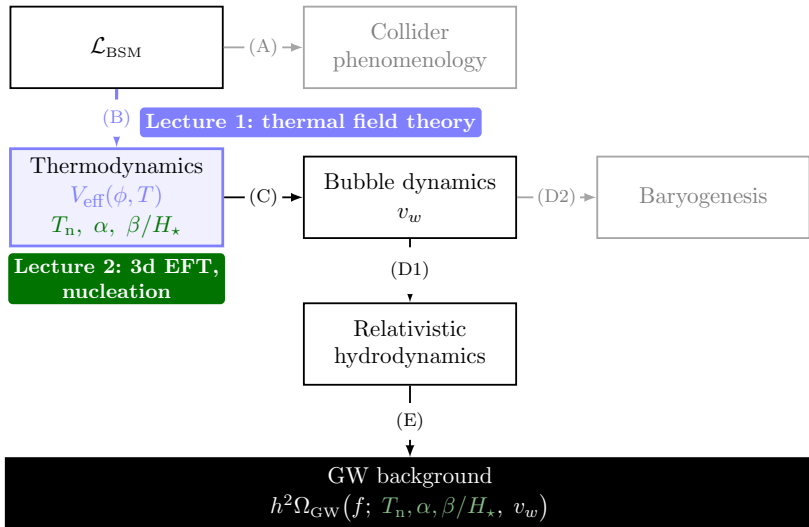
The gravitational-wave pipeline, end to end



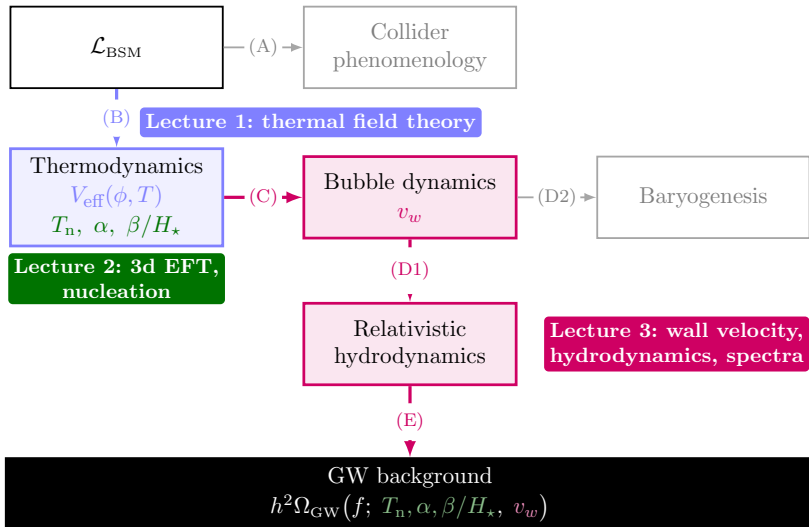
The gravitational-wave pipeline, end to end



The gravitational-wave pipeline, end to end



The gravitational-wave pipeline, end to end



Problem

Problem 4: Show that causality forces $\Omega_{\text{GW}} \propto k^3$ at small k , for any source that is uncorrelated beyond a length ℓ_{B} .

Summary of lecture 3

1: The wall reaches a **terminal velocity** when the driving pressure balances friction and backreaction, $\mathcal{P}(v_w) = 0$. LTE gives an upper bound, the ballistic limit a lower one.

2: Computing v_w means solving the scalar equation of motion, hydrodynamic matching *and* a linearized Boltzmann equation, iteratively, as implemented in **WAGGO**.

3: The spectrum is a broken power law: k^3 in the infrared by causality, and a source-dependent tail. v_w sets both the amplitude, via $\kappa(v_w, \alpha)$, and the peak frequency, via ℓ_B .

The weakest links remain T_n and v_w .

Backup

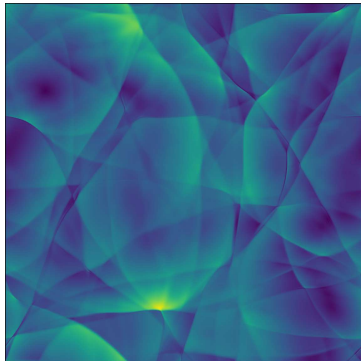
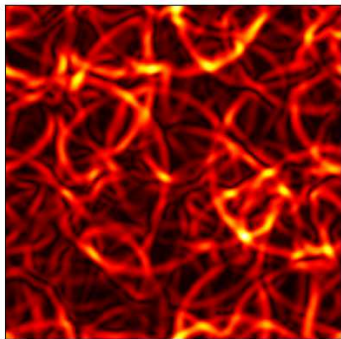
Scales: why LISA

- ▶ Thermal gravitons today: $f \sim T_{\text{now}}/2\pi\hbar \sim 100$ GHz.
- ▶ The wavelength from a transition is at most the horizon, $\ell_{\text{H}} \sim M_{\text{Pl}}/T_{\text{EW}}^2$; redshifted to today, $f_{\ell_{\text{H}}} |_{\text{EW}} \sim 10^{-7}$ Hz, and $f > 10^{-7}$ Hz by causality.
- ▶ LISA armlength 3×10^9 m $\Rightarrow f|_{\text{LISA}} \sim (10^{-5} \dots 10^{-1})$ Hz.
- ▶ Amplitude: the rate is quadratic in suppressing factors,

$$\Omega_{\text{GW}} h^2 \lesssim 10^{-5} \left(\frac{L}{e} \right)^2 \left(\frac{\ell_{\text{B}}}{\ell_{\text{H}}} \right)^2,$$

which with $L/e \sim \ell_{\text{B}}/\ell_{\text{H}} \sim 10^{-2}$ gives $\sim 10^{-13}$, right at the LISA sensitivity.

Simulations do the non-linear part



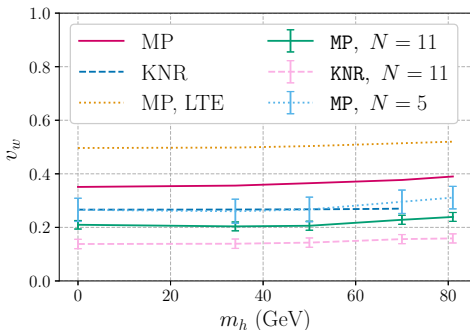
Once bubbles collide there are no symmetries left and hydrodynamics is highly non-linear. Large-scale simulations (and “higgsless” approximations to them) provide the templates; semi-analytic models such as the *sound-shell model* interpolate.

WallGo: the Standard Model with a light Higgs

Comparison with the earlier literature:²⁰ **spectral method** ($N = 11$) against **moment expansion** ($N = 3$).

The differences come from

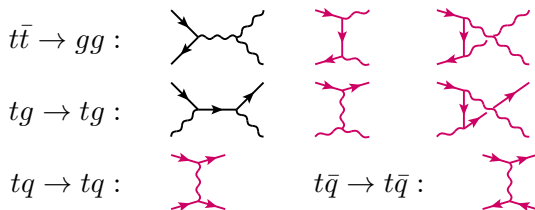
- ▷ the matrix elements in $\mathcal{C}_{ab}$;
- ▷ mixing in the Boltzmann equations, e.g. δf_{top} depends on δf_W .



²⁰**MP**: Moore, Prokopec, *How fast can the wall move? A Study of the electroweak phase transition dynamics*, Phys. Rev. D **52** (1995) 7182 [hep-ph/9506475]; **KNR**: Konstandin, Nardini, Rues, *From Boltzmann equations to steady wall velocities*, JCAP **09** (2014) 028 [1407.3132]

The processes in the collision term

All $2 \rightarrow 2$ scatterings of the top quark with the plasma at order g_s^4 , as generated by WallGoMatrix:²¹

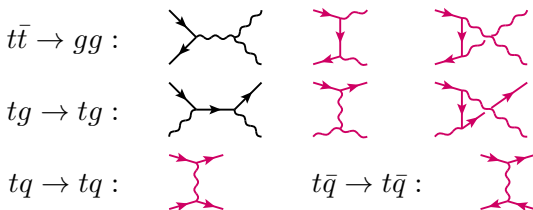


($q = u, d, s, c, b$; an out-of-equilibrium W adds its own list.)

²¹ Ekstedt, Gould, Hirvonen *et al.*, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04** (2025) 101 [2411.04970]

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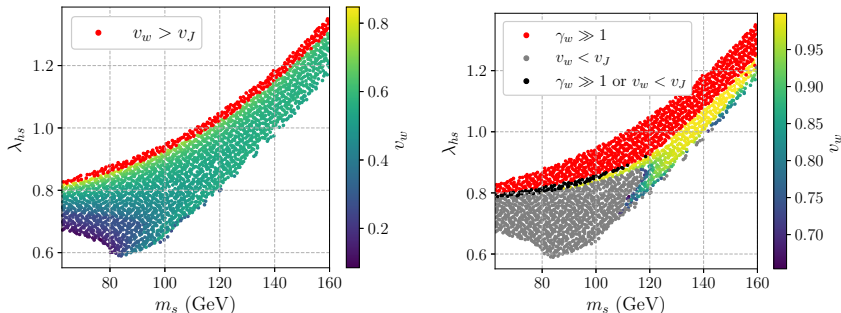
Leading log: only the red diagrams, with a soft exchanged particle, cut off by thermal masses $\sim g_s T$: $\mathcal{C} \propto g_s^4 T \ln(1/g_s)$.²²

²¹ Ekstedt, Gould, Hirvonen *et al.*, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04** (2025) 101 [2411.04970]

²² Arnold, Moore, Yaffe, *Transport coefficients in high temperature gauge theories. 1. Leading log results*, JHEP **11** (2000) 001 [hep-ph/0010177]

Scanning the xSM parameter space

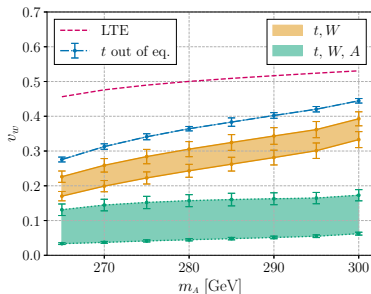
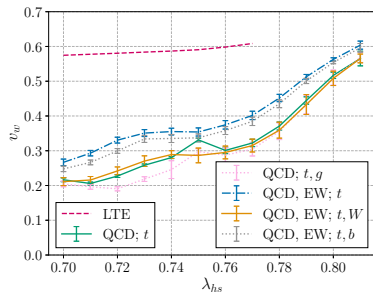
Wall velocity across the xSM parameter space at $\lambda_s = 1$: deflagration (left) and detonation (right) solutions.²³



²³ Ekstedt, Gould, Hirvonen *et al.*, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04** (2025) 101 [2411.04970]

Theoretical uncertainties

$\Delta v_w \approx \Delta \mathcal{P} (d\mathcal{P}/dv_w)^{-1}$: near the zero crossing the slope is small, so a modest uncertainty $\Delta \mathcal{P}$ in the pressure becomes a large Δv_w .²⁴

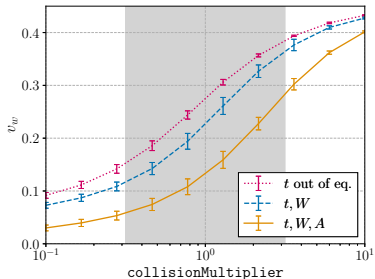
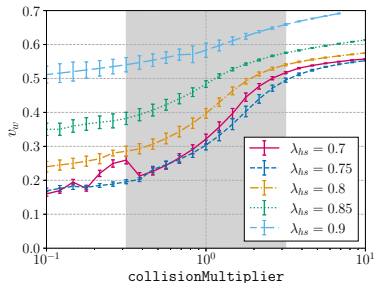


Which species are kept *out of equilibrium* (xSM left, IDM right).

²⁴ van de Vis, Schicho, Niemi, Laurent, Hirvonen, Gould, *WallGo investigates: Theoretical uncertainties in the bubble wall velocity*, JHEP **04** (2026) 041 [2510.27691]

Theoretical uncertainties

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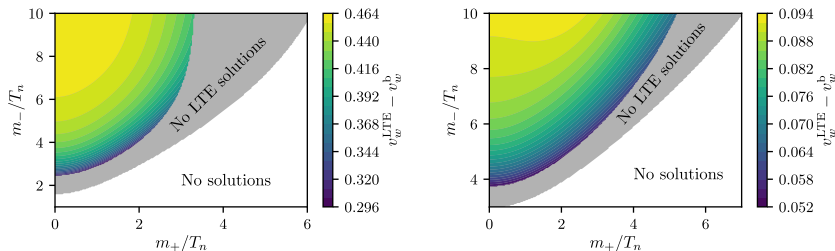


The treatment of the *collision integrals* (xSM left, IDM right).

²⁴ van de Vis, Schicho, Niemi, Laurent, Hirvonen, Gould, *WallGo investigates: Theoretical uncertainties in the bubble wall velocity*, JHEP **04** (2026) 041 [2510.27691]

The gap between the two limits

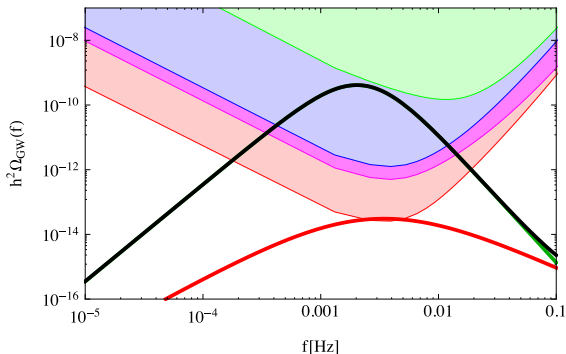
The gap $\Delta v_w = v_w^{\text{LTE}} - v_w^{\text{b}}$ across the xSM, for $\alpha_p = 0.01$ (left) and $\alpha_p = 1$ (right).²⁵



Where the gap is wide, the Boltzmann equation really has to be solved.

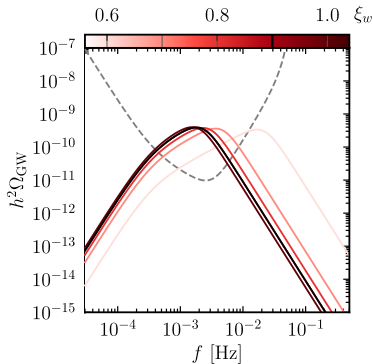
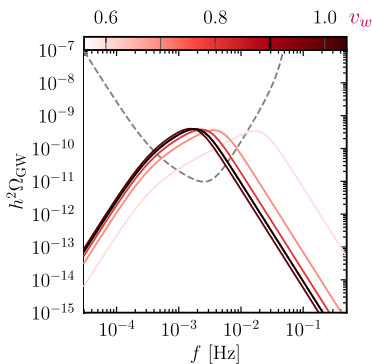
²⁵ Ai, Laurent, van de Vis, *Bounds on the bubble wall velocity*, JHEP **02** (2025) 119 [2411.13641]

From $(T_n, \alpha, \beta/H_\star, v_w)$ to a signal



A benchmark: $\alpha = 0.5$, $\beta/H_\star = 100$, $T_\star = 100$ GeV, $v_w = 0.95$, against the LISA sensitivity curve.

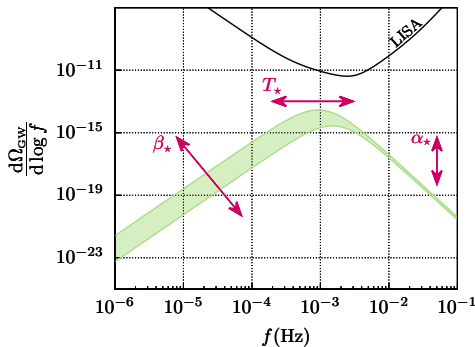
The wall velocity moves the signal



Both the peak amplitude and the peak frequency depend on v_w :

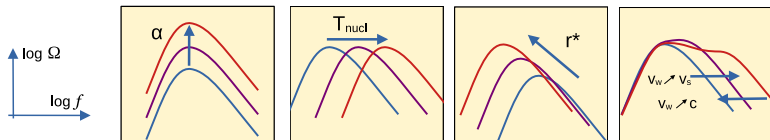
- ▷ amplitude, through the efficiency factor $\kappa(v_w, \alpha)$;
- ▷ frequency, through the mean bubble separation $\ell_B \simeq (8\pi)^{1/3} v_w / \beta$.

What can be reconstructed



Given a detection, the transition parameters can be reconstructed from the spectrum. This works only if the theory uncertainties of lectures 1–3 are smaller than the experimental ones.

What can be reconstructed



The “fingerprint”: how each of T_n , α , $\beta/H_\star$ and v_w moves the spectrum. Degeneracies between them limit the reconstruction.

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